



Thaneeshga M 251801354-2025-AIDS ▾

T2**Started on** Thursday, 24 September 2026, 10:27 AM**State** Finished**Completed on** Thursday, 24 September 2026, 10:37 AM**Time taken** 10 mins 26 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1 | #include<stdio.h>
2 | int main(){
3 |     int n;
4 |     scanf("%d",&n);
5 |     int arr[n];
6 |     for(int i=0;i<n;i++){
7 |         scanf("%d",&arr[i]);
8 |     }
9 |     for(int i=0;i<n;i++){
10 |         for(int j=i+1;j<n;j++){
11 |             if(arr[i]==arr[j]){
12 |                 printf("%d",arr[i]);
13 |                 return 0;
14 |             }
15 |         }
16 |     }
17 |     return 0;
18 | }
```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



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T2**Started on** Thursday, 24 September 2026, 10:38 AM**State** Finished**Completed on** Thursday, 24 September 2026, 11:22 AM**Time taken** 44 mins 8 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n;
5      scanf("%d", &n);
6
7      int arr[n];
8
9      for (int i = 0; i < n; i++)
10         scanf("%d", &arr[i]);
11
12     int slow = arr[0];
13     int fast = arr[0];
14
15     do {
16         slow = arr[slow];
17         fast = arr[arr[fast]];
18     } while (slow != fast);
19
20     slow = arr[0];
21
22     while (slow != fast) {
23         slow = arr[slow];
24         fast = arr[fast];
25     }
26
27     printf("%d", slow);
28
29     return 0;
30 }
31

```

	Input	Expected	Got	
✓	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✓
✓	5 1 2 3 4 4	4	4	✓
✓	5 1 1 2 3 4	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)

**Thaneeshga M 251801354-2025-AIDS** ▾**T2****Started on** Thursday, 24 September 2026, 11:26 AM**State** Finished**Completed on** Thursday, 24 September 2026, 11:27 AM**Time taken** 1 min 54 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2
3 int main() {
4     int T;
5     scanf("%d", &T);
6
7     while (T--) {
8         int n1, n2;
9
10        scanf("%d", &n1);
11        int a[n1];
12
13        for (int i = 0; i < n1; i++)
14            scanf("%d", &a[i]);
15
16        scanf("%d", &n2);
17        int b[n2];
18
19        for (int i = 0; i < n2; i++)
20            scanf("%d", &b[i]);
21
22        for (int i = 0; i < n1; i++) {

```

```
23     for (int j = 0; j < n2; j++) {
24         if (a[i] == b[j]) {
25             printf("%d ", a[i]);
26             break;
27         }
28     }
29 }
30
31 printf("\n");
32 }
33
34 return 0;
35 }
36
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

Back to Course



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T2**Started on** Thursday, 24 September 2026, 11:28 AM**State** Finished**Completed on** Thursday, 24 September 2026, 11:29 AM**Time taken** 1 min 15 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```

1 #include <stdio.h>
2
3 int main() {
4     int T;
5     scanf("%d", &T);
6
7     while (T--) {
8         int m, n;
9
10        scanf("%d", &m);
11        int a[m];
12
13        for (int i = 0; i < m; i++)
14            scanf("%d", &a[i]);
15
16        scanf("%d", &n);
17        int b[n];
18
19        for (int i = 0; i < n; i++)
20            scanf("%d", &b[i]);
21
22        int i = 0, j = 0;
```

```
23
24 while (i < m && j < n) {
25     if (a[i] == b[j]) {
26         printf("%d ", a[i]);
27         i++;
28         j++;
29     }
30     else if (a[i] < b[j]) {
31         i++;
32     }
33     else {
34         j++;
35     }
36 }
37
38 printf("\n");
39 }
40
41 return 0;
42 }
```

	Input	Expected	Got	
✓	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✓
✓	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)



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T2**Started on** Thursday, 24 September 2026, 11:30 AM**State** Finished**Completed on** Thursday, 24 September 2026, 11:31 AM**Time taken** 1 min 20 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n, k;
5      scanf("%d", &n);
6
7      int a[n];
8
9      for (int i = 0; i < n; i++)
10         scanf("%d", &a[i]);
11
12     scanf("%d", &k);
13
14     int i = 0, j = 1;
15
16     while (i < n && j < n) {
17         if (i == j) {
18             j++;
19         }
20         else if (a[j] - a[i] == k) {
21             printf("1");
22             return 0;
23         }
24         else if (a[j] - a[i] < k) {
25             j++;
26         }
27         else {
28             i++;
29         }
30     }
31
32     printf("0");
33
34     return 0;
35 }
36

```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)

**Thaneeshga M 251801354-2025-AIDS** ▾**T2****Started on** Thursday, 24 September 2026, 11:31 AM**State** Finished**Completed on** Thursday, 24 September 2026, 11:32 AM**Time taken** 1 min 15 secs**Marks** 1.00/1.00**Grade** **10.00** out of 10.00 (**100%**)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n, k;
5      scanf("%d", &n);
6
7      int A[n];
8
9      for (int i = 0; i < n; i++) {
10         scanf("%d", &A[i]);
11     }
12
13     scanf("%d", &k);
14
15     int i = 0, j = 1;
16
17     while (i < n && j < n) {
18         if (i == j) {
19             j++;
20             continue;
21         }
22
23         int diff = A[j] - A[i];
24
25         if (diff == k) {
26             printf("1");
27             return 0;
28         }
29         else if (diff < k) {
30             j++;
31         }
32         else {
33             i++;
34         }
35     }
36
37     printf("0");
38
39     return 0;

```

```
40 | j
41 |
```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)